kata – Describe katas with pictograms

Lukas Heindl
oss.heindl+latex@protonmail.com
Github: https://github.com/atticus-sullivan/kata/

0.1.0 from 2026/03/23

The `kata` package provides a compact and structured way to typeset martial arts katas[1] (forms) in LaTeX. It models a kata as a sequence of steps, each combining the stance hand/foot technique, and transition direction, and supports customizing the resulting pictograms for clear visual representation.

Requires: KOMA-documentclass

1 Introduction

In many martial arts such as Karate, a kata is a predefined sequence of movements [2].

The `kata` package aims to provide a simple, structured way to describe katas in LaTeX. Each step of a kata is expressed as a step consisting of

1. the stance with direction
2. a technique with direction
3. the direction of movement or rotation into the next step

The package also offers customizing the resulting pictograms, allowing visual annotation of sequences, such as highlighting repetitions or emphasizing specific actions.

1.1 Simple Example

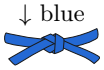
Note this code is "run" after the box showing the code. Because of the use of `\marginline` it was not possible to wrap this in `dispExample*`. Also this way the `\section` command will leak and insert a new section to this documentation. <*doc>

[1] https://en.wikipedia.org/wiki/Kata
[2] https://en.wikipedia.org/wiki/Kata

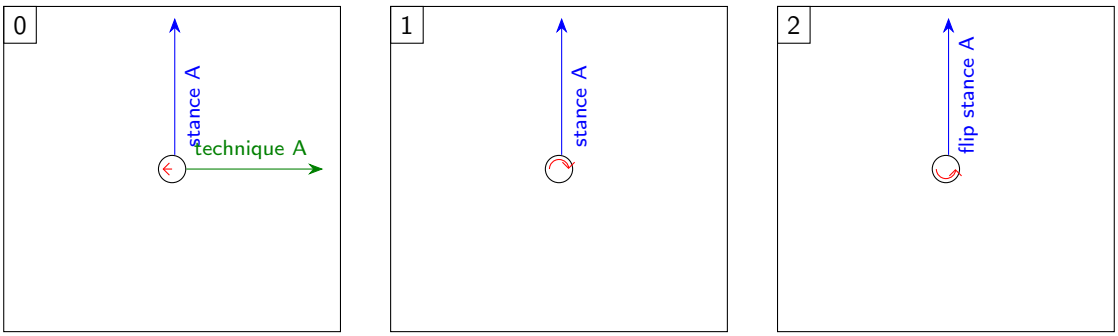
```
\begin{kata}[width=3]{Simple Example}{blue}
  Some explanation -- normal \LaTeX code
  \parindent=0pt

  \begin{step}
    \stance{stance A}{0}
    \waza{technique A}{-90}
    \dir{center}{90}
  \end{step}
  \begin{step}
    \stance{stance A}{0}
    \dir{90}{-90}
  \end{step}
  \begin{step}
    \stance[flip]{stance A}{0}
    \dir[flip]{90}{-90}
  \end{step}
\end{kata}
```

2 Simple Example

↓ blue



Some explanation – normal LaTeXcode



</doc>

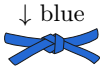
2.1 Extended Example

Note this code is "run" after the box showing the code. Because of the use of `\marginline` it was not possible to wrap this in `dispExample*`. Also this way the `\section` command will leak and insert a new section to this documentation. <*doc>

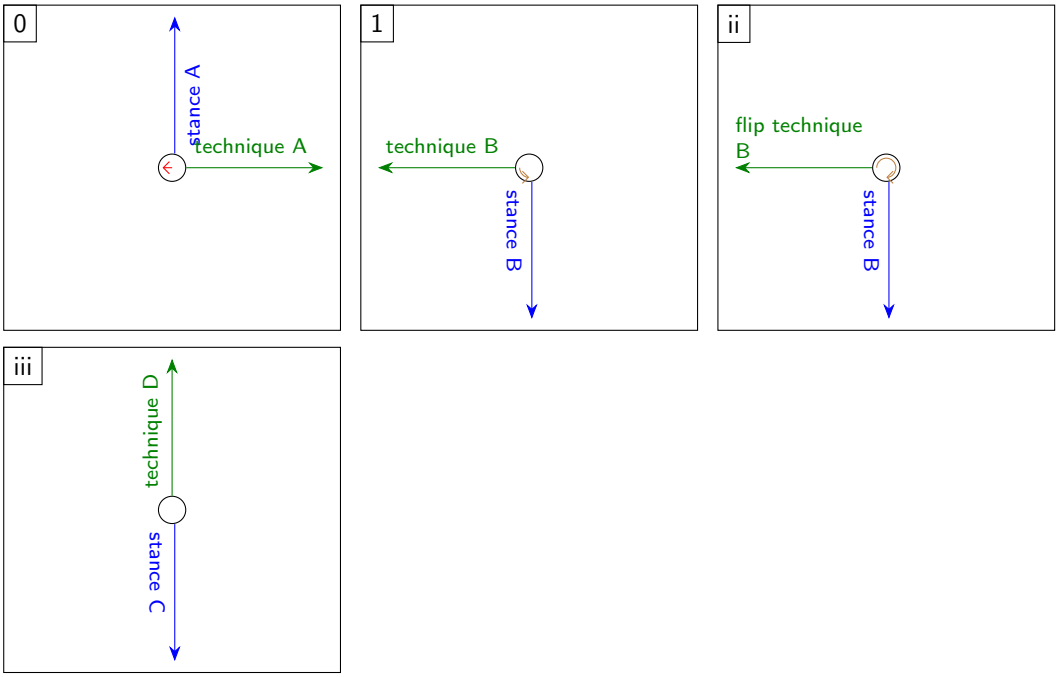
```latex
\begin{kata}[width=3]{Extended Example}{blue}
  Some explanation -- normal \LaTeX code
  \parindent=0pt

  \pgfkeys{/kata/color/waza=green!50!black}
  \begin{step}
    \stance{stance A}{0}
    \waza{technique A}{-90}
    \dir{center}{90}
  \end{step}
  \pgfkeys{/kata/color/dir=brown}
  \begin{step}
    \stance{stance B}{180}
    \waza{technique B}{90}
    \dir{90}{180}
  \end{step}
  \renewcommand\kataStepCnt[1]{\roman{#1}}
  \begin{step}
    \stance{stance B}{180}
    \waza[flip]{technique B}{90}
    \dir[flip]{90}{180}
  \end{step}
  \pgfkeys{
    /kata/customize/stepCnt/.style={xshift=1em},
    /kata/customize/border/.style={/kata/border/arrowUD=45},
  }
  \begin{step}
    \stance{stance C}{180}
    \waza{technique D}{0}
  \end{step}
\end{kata}
```

3 Example

↓ blue



Some explanation – normal LaTeXcode



</doc>

4 Implementation

```
\RequirePackage{memoize}
\RequirePackage{pgfkeys}
\RequirePackage{pgfopts}
\RequirePackage{coloredbelts}
\RequirePackage{tikz}
\usetikzlibrary{calc,positioning,fit,arrows.meta}
\usetikzlibrary{decorations.markings,decorations.pathmorphing,backgrounds}

\RequirePackage{scrbase}
\AtBeginDocument{%
  \@ifundefined{KOMAClassName}{%
    \PackageError{mypackage}
      {This package requires a KOMA-Script class}
      {Use scrartcl, scrreprt, or scrbook.}%
  }{}
}
```

Define marginline in case it isn't already defined (so the package works without KOMA-classes as well).

```
\AddToHook {begindocument} [.] {%
  \providecommand\marginline[1]{%
    \marginpar[{\raggedleft #1}]{\raggedright #1}%
  }%
}
```

4.1 General pgfkeys setup

```
\pgfkeys{
```

for package wide settings

```
  /kata/setup/.is family,
```

for settings regarding for the kata env

```
  /kata/kata/.is family,
```

for settings regarding for the step env

```
  /kata/step/.is family,
```

for settings regarding for the stance command

```
  /kata/stance/.is family,
```

for settings regarding for the waza command

```
  /kata/waza/.is family,
```

for settings regarding for the dir command

```
  /kata/dir/.is family,
```

for settings regarding coloring

```
  /kata/color/.is family,
```

internal style definitions

```
  /kata/style/.is family,
}
```

4.2 Package wide options

Defer keys in setup to the respective locations

```
\pgfkeys{
  /kata/setup,
  width/.style = {/kata/kata,width={#1}},
  color/.style = {/kata/color,#1},
  style/.style = {/kata/style,#1},
  kata/.style = {/kata/kata,#1},
  step/.style = {/kata/step,#1},
  stance/.style = {/kata/stance,#1},
  waza/.style = {/kata/waza,#1},
  dir/.style = {/kata/dir,#1},
  head style/.style = {/kata/style/head/.append style = {#1}},
  border style/.style = {/kata/style/border/.append style = {#1}},
  step counter style/.style = {/kata/style/step counter/.append style = {#1}},
}
```

Setup package option parsing

```
\ProcessPgfPackageOptions{/kata/setup}
```

A setup command for setting package options later

`\kataSetup`

```
\NewDocumentCommand{\kataSetup}{m}{%
  \pgfkeys{/kata/setup,#1}%
  \ignorespaces%
}%
```

4.3 Core internals

A counter for each pictogram to enable referencing.

```
\newcounter{kata@step}[section]
```

`kata@requirestep` enforces something is run within the step environment (helper)

```
\newif\ifkata@instep
\NewDocumentCommand{\kata@requirestep}{m}{%
  \ifkata@instep\else
    \PackageError{kata}{#1 used outside step environment}{}
  \fi
}
```

`kata@requirekata` enforces something is run within the kata environment (helper)

```
\newif\ifkata@inkata
\NewDocumentCommand{\kata@requirekata}{m}{%
  \ifkata@inkata\else
    \PackageError{kata}{#1 used outside kata environment}{}
  \fi
}
```

TODO: is this needed? Or should the keys just set a style instead of setting a conditional?

```
\newif\ifkata@stance@flip
\newif\ifkata@waza@flip
\newif\ifkata@dir@flip
\newif\ifkata@step@first
\newif\ifkata@step@phantom
```

4.4 The kata environment

```
\pgfkeys{
  /kata/kata,
  width/.store in = \kata@width,
  width/.initial = 6,
  width/.value required,
```

defer keys to the respective locations

```
  step/.style = {/kata/step,#1},
  stance/.style = {/kata/stance,#1},
  waza/.style = {/kata/waza,#1},
  dir/.style = {/kata/dir,#1},
```

allow directly appending to styles

```
  head style/.style = {/kata/style/head/.append style = {#1}},
  border style/.style = {/kata/style/border/.append style = {#1}},
  step counter style/.style = {/kata/style/step counter/.append style = {#1}},
}
```

`kata` (*env.*) this environment is basically a container for a sequence of steps. Arguments:

1. optional: keys from `/kata/kata`
2. kata name
3. for which belt / marginline
4. optional: belt color (for drawing), if empty -> not drawn, if not present use arg 3

```
\NewDocumentEnvironment{kata}{O{}mmo}{%
  \pgfkeys{/kata/kata,#1}
  \section{#2}
  \Ifstr{#3}{}{
```

show if arg 3 is empty but arg 4 was set

```
    \IfNoValueF{#4}{\kata@beltMargin{}{#4}}%
  }{%
```

use arg 3 as fallback if arg 4 was not set

```
    \kata@beltMargin{#3}{\IfNoValueTF{#4}{#3}{#4}}%
  }%
  \setlength{\lineskip}{6pt}\noindent%
  \setcounter{kata@step}{0}%
  \kata@inkatatrue%
  \ignorespaces%
```

there is no harm in a single word/pictogram in the last line to turn off the corresponding penalty

```
  \widowpenalty=0%
```

allow large larger spaces between the pictograms if this avoids drawing a picture outside of the textarea

```
  \sloppy
}{%
```

add hidden steps to the last line. This ensures a uniform looking grid of pictograms

```
  \pgfmathtruncatemacro{\kata@missing}{mod(\kata@width - mod(\c@kata@step,\kata@width),\kata@wid
  \setcounter{kata@step}{0}%
  \ifnum\kata@missing>0%
    \foreach \i in {1,...,\kata@missing} {%
      \begin{step}[phantom]\end{step} %
    }%
  \fi%
  \kata@inkatafalse%
```

avoid left-aligning the last line. Instead it should be justified in order to obtain a proper grid

```
  \parfillskip=0pt%
  \par%
}
```

`\kata@beltMargin`

```
\NewDocumentCommand{\kata@beltMargin}{mm}{%
```

```
    \marginline{%
      \centering\footnotesize%
      \Ifstr{#1}{}{}{%
        $\downarrow$ #1%
      }%
```

only linebreak if both are non-empty

```
      \Ifstr{#1}{}{}{\Ifstr{#2}{}{}{\\}}%
      \Ifstr{#2}{}{}{%
        \ColorBelt*[scale=0.1]{#2}%
      }%
    }%
  }
```

4.5 The step environment

```
  \pgfkeys{
    /kata/step,
    first step/.is if = kata@step@first,
    first step/.default = true,
    phantom/.is if = kata@step@phantom,
    phantom/.default = true,
    phantom = false,
    phantom/.append code={\ifkata@step@phantom\pgfkeysalso{/kata/style/step phantom}\fi},
    phantom/.style = {/kata/style/step phantom},
```

allow directly appending to styles

```
    style/.style = {/kata/style/step/.append style = {#1}},
    first step style/.style = {/kata/style/step first/.append style = {#1}},
    phantom style/.style = {/kata/style/step phantom/.append style = {#1}},
```

defer keys to the respective locations

```
    stance/.style = {/kata/stance,#1},
    waza/.style = {/kata/waza,#1},
    dir/.style = {/kata/dir,#1},
  }
```

`step` (*env.*) this environment always should wrap `kata@step`. It handles the aspects of positioning the pictogram. Arguments: 1 (optional): whether this is the first step of the kata

```
  \NewDocumentEnvironment{step}{O{}+b}{%
    \kata@requirekata{\string\step}%
    \pgfkeys{/kata/step,#1}%
    \ifkata@step@first\par\nointerlineskip\vfil\else\fi%
    \begin{kata@step}%
      #2
    \end{kata@step}%
    \ifkata@step@phantom\else\refstepcounter{kata@step}\fi%
  }{}
```

`kata@step` (*env.*) the environment to draw everything else in

```
  \NewDocumentEnvironment{kata@step}{}{%
```

```
    \kata@requirekata{\string\step}%
    \begin{tikzpicture}[font=\scriptsize]%
      \begin{scope}[local bounding box=output]%
        \kata@head\kata@nothing%
        \kata@insteptrue%
  }{
      \kata@instepfalse%
      \end{scope}%
```

Drawing the border on the background has the advantage the rectangle can be filled and colorize the background as well.

```
      \begin{scope}[on background layer]%
        \node[/kata/style/border] (t1) {};%
        \node[/kata/style/step counter] (cnt) {\kataStepCnt{kata@step}};%
      \end{scope}%
    \end{tikzpicture}%
  }
```

4.6 Specify the Content

`\kiai` draw indicator for *kiai*. Arguments:

1. (optional) distance to head
2. direction

```
  \NewDocumentCommand{\kiai}{O{}m}{
    \kata@requirestep{\string\kiai}
    \def\base{#2+90} \def\off{30} \def\ext{40} \def\rOff{4pt} \def\len{0.75}
    \draw[shorten <=10pt,CtpRed]
    ($(head)+(\base-\off:\rOff)$) edge[decorate,decoration={bent,aspect=0.3,amplitude=-2.5pt}] coo
    ($(head)+(\base+\off:\rOff)$) edge[decorate,decoration={bent,aspect=0.3,amplitude=+2.5pt}] coo
    \path (tmpA) -- (tmpB) node[midway,sloped,font={\sffamily\scriptsize},CtpRed,draw,decorate,dec
  }
```

4.6.1 The stance command

```
  \pgfkeys{
    /kata/stance,
    flip/.is if = kata@stance@flip,
    flip/.default = true,
    style/.style = {/kata/style/stance/.append style = {#1}},
    label/.style = {/kata/style/stance label/.append style = {#1}},
    label flipped/.style = {/kata/style/stance label flipped/.append style = {#1}},
  }
```

`\stance` draw the arrow/path showing the stance Usage:

1. (optional) side to use (L/R)
2. name/text

3. direction

```
\NewDocumentCommand{\stance}{O{}mmO{}}{%
  \pgfkeys{/kata/stance,#4}%
  \kata@requirestep{\string\stance}%
  \pgfmathsetmacro{\kata@dir}{#3+90}%
  \pgfmathsetmacro{\x}{abs(sin(\kata@dir))}%
  \pgfmathsetmacro{\y}{-abs(cos(\kata@dir))}%
  \IfNoValueTF{#1}{\def\kata@nodeContent{}}{\def\kata@nodeContent{#1 }}%
  \draw[/kata/style/stance={\x}{\y}]
    (head) -- ++(\kata@dir:2)
    node[\ifkata@stance@flip/kata/style/stance label flipped\else/kata/style/stance label\fi]
    {\hangafter=1\settowidth\hangindent{\kata@nodeContent}\kata@nodeContent#2}
  ;
}
```

4.6.2 The waza command

```
\pgfkeys{
  /kata/waza,
  flip/.is if = kata@waza@flip,
  flip/.default = true,
  style/.style = {/kata/style/waza/.append style = {#1}},
  label/.style = {/kata/style/waza label/.append style = {#1}},
  label flipped/.style = {/kata/style/waza label flipped/.append style = {#1}},
}
```

`\waza` draw the arrow/path showing a technique. Arguments:

1. (optional) side to use (L/R)
2. name/text
3. direction

```
\NewDocumentCommand{\waza}{O{}mmO{}}{%
  \kata@requirestep{\string\waza}%
  \pgfkeys{/kata/waza,#4}%
  \IfNoValueTF{#1}{\def\kata@nodeContent{}}{\def\kata@nodeContent{#1 }}%
  \pgfmathsetmacro{\kata@dir}{#3+90}%
  \draw[/kata/style/waza]
    (head) -- ++(\kata@dir:2)
    node[\ifkata@waza@flip/kata/style/waza label flipped\else/kata/style/waza label\fi]
    {\hangafter=1\settowidth\hangindent{\kata@nodeContent}\kata@nodeContent#2}
  ;
}
```

4.6.3 The dir command

```
\pgfkeys{
  /kata/dir,
```

```
  flip/.is if = kata@dir@flip,
  flip/.default = true,
  style/.style = {/kata/style/dir/.append style = {#1}},
}
```

`\dir` draw a short arrow/path indicating the next turn/direction. Arguments:

1. (optional) flip
2. source (Center,angle)
3. dst (angle)

```
\NewDocumentCommand{\dir}{O{}mm}{
  \kata@requirestep{\string\dir}
  \pgfkeys{/kata/dir,#1}
  \pgfmathsetmacro{\kata@out}{#3+90}
  \Ifstr{#2}{center}{
    \draw[/kata/style/dir] (head.center) -- ++(\kata@out:.33em);
  }{
    \pgfmathsetmacro{\kata@in}{#2+90}
    \pgfmathsetmacro{\kata@outDeltaI}{\kata@out-\kata@in}
    \ifkata@dir@flip
      \if -\kata@outDeltaI
        \pgfmathsetmacro{\kata@outDeltaII}{360+\kata@outDeltaI}
      \else
        \pgfmathsetmacro{\kata@outDeltaII}{\kata@outDeltaI-360}
      \fi
    \else
      \pgfmathsetmacro{\kata@outDeltaII}{\kata@outDeltaI}
    \fi
    \draw[/kata/style/dir]
      (head.center) ++ (\kata@in:.33em)
      arc[start angle=\kata@in,delta angle=\kata@outDeltaII,radius=.33em]
    ;
  }
}
```

4.6.4 Internal elements

`\kata@nothing` ensure the box is always at least the same size (internal command + always placed)

```
\NewDocumentCommand{\kata@nothing}{}{
  \draw[draw=none] (head) -- ++(0:2.15) -- ++(180:4.3) -- ++(90:2.15) -- ++(-90:4.3);
}
```

`\kata@head` draw the circle symbolizing the head (internal command + always placed)

```
\NewDocumentCommand{\kata@head}{}{
  \node[/kata/style/head] (head) {};
}
```

4.7 Color keys

```
\pgfkeys{
  /kata/color,
  %
  stance/.store in = \kata@color@stance,
  stance/.initial = blue,
  stance = blue,
  %
  waza/.store in = \kata@color@waza,
  waza/.initial = green!50!black,
  waza = green!50!black,
  %
  dir/.store in = \kata@color@dir,
  dir/.initial = red,
  dir = red,
  %
  head/.store in = \kata@color@head,
  head/.initial = black,
  head = black,
  %
  border/.store in = \kata@color@border,
  border/.initial = black,
  border = black,
  %
  step counter/.store in = \kata@color@stepcounter,
  step counter/.initial = black,
  step counter = black,
}
```

4.8 Styles

4.8.1 Styles for drawn elements

```
\pgfkeys{
  /kata/style,
  %
  head/.style = {
    circle,
    draw=\kata@color@head,
    inner xsep=0.333em,
  },
  %
  stance/.style 2 args = {
    draw=\kata@color@stance,
    -{Stealth[scale=1.2]},
    transform canvas={xshift=#1,yshift=#2},
  },
  %
  waza/.style = {
    draw=\kata@color@waza,
```

```
    -{Stealth[scale=1.2]},
  },
  %
  dir/.style = {
    draw=\kata@color@dir,
    -{Straight Barb},
  },
  %
  border/.style = {
    rectangle,
    draw=\kata@color@border,
    fit=(output),
    inner sep=0pt,
    outer sep=0pt,
    inner xsep=0.2em,
  },
  %
  step counter/.style = {
    draw=\kata@color@stepcounter,
    font=\sffamily\small,
    outer sep=0pt,
    at={(t1.north west)},
    below right,
    rectangle,
  },
}
```

4.8.2 Styles for labels

```
\pgfkeys{
  /kata/style,
  %
  stance label base/.style = {
    \kata@color@stance,
    midway,
    sloped,
    font=\sffamily\scriptsize,
    text width=1.8cm,
  },
  stance label/.style = {
    /kata/style/stance label base,
    below,
  },
  %
  stance label flipped/.style = {
    /kata/style/stance label base,
    above,
  },
  %
  waza label base/.style = {
```

```
    \kata@color@waza,
    midway,
    sloped,
    font=\sffamily\scriptsize,
    text width=1.8cm,
  },
  %
  waza label/.style = {
    /kata/style/waza label base,
    above,
  },
  %
  waza label flipped/.style = {
    /kata/style/waza label base,
    below,
  },
}
```

4.8.3 Styles for steps

```
\pgfkeys{
  /kata/style,
  %
  step/.style = {
  },
  %
  step first/.style = {
  },
  %
  step phantom/.style = {
    /kata/style/border/.style={draw=none},
    /kata/style/head/.style={draw=none},
    /kata/style/step counter/.style={opacity=0},
  },
}
```

4.9 Misc

`\kataStepCnt` Formats the counter for the step indicator

```
\NewDocumentCommand{\kataStepCnt}{m}{\arabic{#1}}
```

4.10 Memoization

TODO: which pgfkeys to add TOOD: make optional depending on whether memoize is loaded or not

```
\mmzset{auto={kata@step}{
  memoize,
  csname meaning to context={environment kata@step\space code},
  csname meaning to context={environment kata@step end aux\space},
  csname meaning to context={nothing code},
```

```
  csname meaning to context={head code},
  csname meaning to context={stance code},
  csname meaning to context={waza code},
  csname meaning to context={dir code},
}}
```

Change History

0.1.0
General: Converted to DTX file 1